

# Final Project

## IGME 382 - Maps, Mapping, and Geospatial Technologies

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### Project Overview

For your final project, you will design and complete a small geospatial project on a topic that interests you. You will:

- Propose and investigate a clear spatial question or theme
- Work with at least two data sources
- Modify at least one data source using ArcGIS Pro (desktop)
- Use at least one additional geospatial tool (e.g., StoryMap, ArcGIS Online, Felt, Python, etc.)
- Produce a final geospatial product (e.g. map, app, story) and a written report (4-6 pages)

### Project Proposal

You will first submit a Project Proposal so I can confirm your idea is feasible and meets the requirements.

**Due Date:** Sunday, November 23<sup>rd</sup>, at 11:59PM

Your proposal should be about 1 page and must include:

1. Topic & Question
  - a. The general topic (e.g. park access, flooding risk, food deserts, land cover change)
  - b. Your proposed spatial question or objective in 1-2 sentences
2. Data Sources & Planned Modifications
  - a. List at least two data sources that you plan to use. For each include:
    - i. The name of the dataset and what it represents
    - ii. The URL or source of the dataset
  - b. For at least one of the datasets state how you plan to modify it using ArcGIS Pro (e.g. clip to a study area, buffer, spatial join, dissolve)
    - i. ArcGIS Pro **must** be used for the modification
3. Other Tools (Beyond ArcGIS Pro)
  - a. Identify at least one additional tool you will use (e.g. StoryMaps, Felt, ArcGIS Online, Python/GeoPandas, Mapbox)
  - b. Briefly describe how you plan to use the tool (e.g. analysis, visualization, creating an interactive map)
4. Description of the Final Product

- a. A short paragraph describing what you expect your final deliverable to look like. It should be clear how the final deliverable meets the given requirements
- b. Example:
  - i. A StoryMap with 2 web maps and 3 sections explaining park access in X City

You are allowed to adjust your plan later, but substantial changes should be discussed with me so your project still meets all requirements.

## Project Question or Theme

Your project must be built around a specific spatial question or theme, such as:

- “Which neighborhoods in [city] have poor access to parks within a 10-minute walk?”
- “Where are the highest-risk areas for flooding near [river/city]?”
- “How has land cover changed in [region] between 2015 and 2025?”

Your question should be specific enough that mapping and basic analysis can help answer it.

## Data Requirements

You must:

- Use at least two distinct data sources, from different providers (e.g. Census + local open data, Living Atlas + OpenStreetMap)

## Required Use of ArcGIS Pro

You must use ArcGIS Pro to modify at least one of your data sources in a significant way.

**Simply adding to the data or changing symbology is not enough.**

Examples of significant modifications:

- Clip, select, or erase operations
- Buffering points/lines/polygons
- Spatial or attribute joins
- Summarize Within, Dissolve, or other aggregation tools
- Raster analysis or classification

These can be chained as part of a workflow (e.g., clip → buffer → spatial join).

# Required Use of at Least One Other Tool

You must also use at least one additional geospatial tool to perform analysis and/or present your results.

Examples:

- StoryMap
  - At least 5 content blocks/sections (text, maps, images, etc.)
- ArcGIS Online web map / Felt
  - At least 3 layers, custom symbology, and meaningful pop-ups (not default auto-generated clutter)
- Python / GeoPandas / Jupyter
  - Read spatial data, perform at least one spatial operation, and create at least one map/plot

## Report

Your final report must include the information outlined below. You may include screenshots and figures wherever relevant. It is expected to be between 4-6 pages double spaced.

- Topic
  - Your chosen topic and why it matters
  - The final version of your research question/objective
- Data
  - The datasets you used, what they represent, and where they came from
  - The reason you chose the datasets
- Methods
  - Explain the data modification you made in ArcGIS Pro
  - Explain what you did in the other geospatial tool you chose
  - Explain any issues you encountered and how you got around those issues
  - Explain any changes from the original proposed work
    - Note that significant changes should be approved by me ahead of time
- Results
  - Describe the results and key findings
- Discussion
  - Interpret what the results mean
  - Explain any limitations (e.g. data quality, scale, resolution)
  - Explain any relevant ethics or privacy concerns
- Conclusion
  - Explain what you learned
  - Explain possible next steps or improvements
- Citations

- Include references for data sources, and any tutorials, articles, or documentation used
- The citations may be in any format

## Final Deliverables

You must submit:

1. Final geospatial product (e.g. StoryMap, web map, PDF of map, etc)
  - a. You **must** make sure that I can access the final product. If it is a web map, make sure the permissions are set up correctly.
2. Written Report
  - a. 4-6 pages double spaced

# Grading Rubric

The project proposal will be scored as follows:

## Project Proposal (10 points possible)

| Point Range | Description                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9-10        | <ul style="list-style-type: none"><li>• Clearly states topic and focused spatial question</li><li>• Lists 2+ data sources with sources/URLs</li><li>• Includes a realistic and meaningful modification plan</li><li>• Identifies at least one additional tool and a plausible final product description</li><li>• No spelling and grammar mistakes and written using a professional style</li></ul> |
| 6-8         | <ul style="list-style-type: none"><li>• Proposal includes most required elements but is somewhat vague (e.g., incomplete data source info, fuzzy product description) or needs minor clarification</li><li>• Minor spelling and grammar mistakes</li></ul>                                                                                                                                          |
| 0-5         | <ul style="list-style-type: none"><li>• Proposal missing major components, unclear, not feasible, or not submitted</li><li>• Significant spelling, grammar, and style issues</li></ul>                                                                                                                                                                                                              |

The final deliverables will be scored as follows:

**Question and Scope (9 points possible)**

| Point Range | Description                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7-9         | <ul style="list-style-type: none"><li>• Final project has a clear, focused spatial question</li><li>• Scope is appropriate for course level and timeframe</li></ul> |
| 4-6         | <ul style="list-style-type: none"><li>• Question is mostly clear but somewhat broad/vague or only loosely connected to the work</li></ul>                           |
| 0-3         | <ul style="list-style-type: none"><li>• Question unclear, missing, or not reflected in the project</li></ul>                                                        |

**Data Selection and Documentation (8 points possible)**

| Point Range | Description                                                                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7-8         | <ul style="list-style-type: none"><li>• Uses 2+ distinct data sources</li><li>• Data sources and rationale are clearly documented</li></ul>                      |
| 4-6         | <ul style="list-style-type: none"><li>• Uses 2+ distinct data sources</li><li>• Documentation is mostly correct but has minor gaps or weak explanation</li></ul> |
| 0-3         | <ul style="list-style-type: none"><li>• Fewer than 2 data sources</li><li>• Documentation is incomplete or confusing</li></ul>                                   |

### Data Manipulation using ArcGIS Pro (10 points possible)

| Point Range | Description                                                                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8-10        | <ul style="list-style-type: none"><li>● Uses ArcGIS Pro to meaningfully modify at least one data source</li><li>● Workflow is logical and clearly explained</li><li>● Screenshots or evidence included.</li></ul> |
| 4-7         | <ul style="list-style-type: none"><li>● Uses ArcGIS Pro to meaningfully modify at least one data source</li><li>● Description is present but limited</li></ul>                                                    |
| 0-3         | <ul style="list-style-type: none"><li>● ArcGIS Pro not used for a meaningful modification or not used at all</li><li>● Missing description</li></ul>                                                              |

### Use of Second Tool (15 points possible)

| Point Range | Description                                                                                                                                                                                                                                                                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11-15       | <ul style="list-style-type: none"><li>● Additional tool is used in a substantive way<ul style="list-style-type: none"><li>○ e.g. StoryMap with multiple sections, well-configured web map, Python with real analysis</li></ul></li><li>● Clearly adds value beyond ArcGIS Pro alone</li></ul> |
| 5-10        | <ul style="list-style-type: none"><li>● Tool is used but somewhat minimally<ul style="list-style-type: none"><li>○ e.g. basic web map or lightly customized StoryMap</li></ul></li><li>● Some added value but limited depth</li></ul>                                                         |
| 0-4         | <ul style="list-style-type: none"><li>● Tool use is superficial, clearly last-minute, or missing</li></ul>                                                                                                                                                                                    |

### Map / App Design & Cartography (15 points possible)

| Point Range | Description                                                                                                                                                                                                                                      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11-15       | <ul style="list-style-type: none"><li>• Maps/products are clear, legible, and thoughtfully designed</li><li>• Good use of symbology, color, hierarchy, labels, legends, and titles</li><li>• “No data” and caveats are clearly handled</li></ul> |
| 5-10        | <ul style="list-style-type: none"><li>• Maps are mostly readable but have noticeable design issues<ul style="list-style-type: none"><li>◦ e.g. awkward color choices, clutter, missing or confusing legend/labels</li></ul></li></ul>            |
| 0-4         | <ul style="list-style-type: none"><li>• Maps are hard to interpret, misleading, or missing key cartographic elements</li></ul>                                                                                                                   |

### Analysis, Interpretation & Insight (15 points possible)

| Point Range | Description                                                                                                                                                                                     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11-15       | <ul style="list-style-type: none"><li>• Results are accurately interpreted</li><li>• Findings directly address the question</li><li>• Patterns and implications are clearly discussed</li></ul> |
| 5-10        | <ul style="list-style-type: none"><li>• Some interpretation</li><li>• Connections to the question are present but shallow or incomplete</li></ul>                                               |
| 0-4         | <ul style="list-style-type: none"><li>• Little or inaccurate interpretation</li><li>• Weak linkage to the question</li></ul>                                                                    |



### Limitations, Ethics, Next Steps & Reflection (10 points possible)

| Point Range | Description                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7-10        | <ul style="list-style-type: none"><li>● Thoughtful discussion of data limitations and any ethical/privacy concerns</li><li>● Reflects on what could be improved or extended</li></ul> |
| 3-6         | <ul style="list-style-type: none"><li>● Mentions some limitations or concerns but briefly or superficially</li></ul>                                                                  |
| 0-2         | <ul style="list-style-type: none"><li>● Little or no reflection on limitations or ethics</li></ul>                                                                                    |

### Report Quality and Organization (8 points possible)

| Point Range | Description                                                                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6-8         | <ul style="list-style-type: none"><li>● Report is well organized</li><li>● Writing is clear and mostly free of errors</li><li>● Figures have captions and are referenced in the text</li><li>● Sources are cited</li></ul> |
| 3-5         | <ul style="list-style-type: none"><li>● Generally understandable but with some organization and/or grammar issues</li><li>● Figures/captions could be clearer</li></ul>                                                    |
| 0-2         | <ul style="list-style-type: none"><li>● Report is hard to follow</li><li>● Many writing issues</li><li>● Missing or unlabeled figures</li></ul>                                                                            |